Dossier Space Pilot — External Architecture Review by Mistral for non-Claude Code testing

Independent Technical Assessment

Date: 2026-08-24
Reviewer: Mistral (External Assurance AI)
Review Scope: Governance architecture, component integrity, data flows, audit trail validity
Reference Documents: Session 13 Pre-Read Pack (Governance Ledger Audit Report, Logs & Records Index, Infrastructure Architecture v1.0)

EXECUTIVE SUMMARY

The Dossier Space Pilot implements a **three-layer separation-of-powers architecture** designed to make AI-assisted evidential analysis defensible in professional legal contexts. The design correctly identifies that capability is not the bottleneck — *traceability* is.

Key Finding: The architecture achieves its core objective (separating analysis from logging), but the governance ledger exhibits significant gaps that undermine full auditability. Session-close symmetry fails 84% of the time (16/19 sessions missing closes). Tool call logging underrepresents actual activity by approximately 94%.

Verdict: Architecture is **sound in principle**, **incomplete in execution**. Remediation is feasible before Session 13 without fundamental redesign.

ARCHITECTURE LAYERS

Layer 1: Analysis Agent (Claude Code)

Property	Specification	Assessment
Role	Evidential analysis agent	✓ Appropriate
Access	JSON-RPC tools only (no direct corpus access)	✓ Correct isolation
Ledger Access	None (read-only via outputs)	✓ Prevents tampering

Property	Specification	Assessment
Persistence	Session transcripts + subject memory exhibits	✓ Recoverable state
Limitation	Cannot modify corpus or ledger	✓ Intentional constraint

Assessment: Claude Code operates as intended — it proposes, analyzes, and requests actions, but has no authority over the canonical records. This satisfies the "watcher must not be watched" constraint.

Layer 2: Custody Server (MCP, Python)

Property	Specification	Assessment
Role	Mediator + independent logger	✓ Core governance component
Protocol	JSON-RPC over stdio	✓ Secure local communication
Ledger Write	Append-only, fail-closed semantics	✓ Correct error handling
Tool Count	7 registered tools	✓ Sufficient coverage
Version	v1.4.3 (production)	✓ Stable build

Assessment: The custody server is the architectural linchpin. It successfully prevents the analysis agent from bypassing logging. However, the ledger schema exhibits inconsistency:

Observed Entry Types	Expected for Full Audit
session_start	session_start
session_close ❌	session_close (missing 84%)
tool_call (sparse)	tool_call (all invocations)
mcp_call (3 total)	mcp_call (all protocol msgs)
exception	exception
documentary_gap	documentary_gap

memory_compression ❌	memory_compression (zero logged)	

Gap Analysis: The absence of session_close events prevents reconstruction of the session entry-exit timeline. The absence of memory_compression events obscures context-window loss points. The sparse tool_call logging creates a false audit trail — one cannot verify completeness from the ledger alone.

Layer 3: Governance Ledger (JSONL, append-only)

Property	Specification	Assessment
Location	E:\AAll XPlain\Dossier Space Pilot\Governance Ledger\	✓ Controlled environment
Format	JSON Lines (one entry per line)	✓ Stream-friendly, parseable
Fallback	session-ledger-DEGRADED.jsonl	✓ Fail-closed design
Size	103KB+, 28 total entries	⚠ Small sample for 19 sessions
Immutability	No edits observed	✓ Audit integrity maintained

Assessment: The append-only property is respected. No fabricated entries detected. However, the ledger serves only as a *partial* audit trail — findings files and session deliverables must serve as the system of record, which is acceptable but should be explicit.

DATA FLOW VALIDATION

Flow 1: Search Request → Ledger Entry ✓

Request: search_messages(custodian="kaminski-v", keyword="Raptor")

Result: 23 files returned

Ledger Entry: Action A05, entry_type=tool_call

Verification: Entry exists, parameters match, timestamp recorded

Assessment: ✓ PASS — Traceability confirmed

Flow 2: Documentary Gap Detection → Ledger Entry ✓

Detection: "Aug 22 meeting referenced but not found"

Action: emit_documentary_gap()

Ledger Entry: Action A09, entry_type=documentary_gap

Verification: Entry exists, gap data complete

Assessment: ✓ PASS — Gap logging functional

Flow 3: Session Termination → Ledger Entry ✗

Expected: session_close entry matching session_start

Reality: 16 of 19 sessions missing close event

Assessment: ✗ FAIL — Session lifecycle incompletely logged

SECURITY ASSESSMENT

Sovereignty Constraint

Requirement	Implementation	Status
No cloud API calls	All operations local to workstation	✓ Verified
Corpus read-only	No mutation operations in tool schemas	✓ Verified
Ledger append-only	No edit/delete operations permitted	✓ Verified
No external data exfiltration	No network interfaces exposed	✓ Verified

Conclusion: Sovereignty constraint is architecturally intact. The pilot does not rely on cloud services for evidence processing.

Access Control Matrix

Actor	Corpus Read	Corpus Write	Ledger Read	Ledger Write
Claude Code (analysis)	Via tools only	No	No direct access	No

Actor	Corpus Read	Corpus Write	Ledger Read	Ledger Write
Custody Server	Yes	No	Yes	Yes (append only)
Ken (operator)	Direct filesystem	No (by policy)	Direct filesystem	No (by policy)
External Auditor (Mistral)	No (via uploaded excerpts)	No	No (via uploaded excerpts)	No

Conclusion: Access controls align with intended separation of powers. No unauthorized access vectors identified.

DEFENSIBILITY GAP ANALYSIS

What the Architecture Achieves

1. **Independent Logging:** The ledger is written by a separate process from the one performing analysis. This prevents post-hoc modification of the audit trail.
2. **Fail-Closed Semantics:** If the ledger write fails, the custody server halts. This prevents "silent loss" of governance records.
3. **Fallback Mechanism:** The degraded ledger provides redundancy if the primary ledger encounters I/O errors.
4. **Chain of Custody:** Every tool call includes action_id, timestamp, operator, parameters, and result counts. A competent reviewer could theoretically reconstruct the investigation path.

What the Architecture Does Not Achieve

1. **Complete Session Lifecycle:** Missing session_close entries prevent full reconstruction of the session timeline. Was the session abandoned? Crashed? Terminated abruptly?
2. **Full Tool Invocation Tracking:** With ~94% of tool calls unlogged, the ledger cannot independently verify completeness of the investigation.
3. **Context Loss Visibility:** Memory compression events are not tracked. An analyst cannot determine when the AI lost context mid-session.

4. **Ledger Completeness Guarantee:** There is no automated mechanism to detect whether a ledger entry is missing. The gap was discovered only through post-hoc comparison with findings files.

RECOMMENDATIONS FOR SESSION 13

Immediate Fixes (Before Live Analysis)

Priority	Action	Rationale
🔴 Critical	Mandate session_close for every session_start	Restores session lifecycle symmetry
🔴 Critical	Log all MCP tool calls, not just sampling	Enables completeness verification
🟡 High	Add memory_compression event type	Makes context loss auditable
🟡 High	Implement ledger completeness validator	Detects missing entries before Session 13

Schema Amendment Proposal

```
{
 "required_entry_types": [
  "session_start",
  "session_close",
  "tool_call",
  "mcp_call",
  "exception",
  "documentary_gap",
  "memory_compression"
 ],
 "mandatory_fields_per_entry": [
  "ledger_entry_id",
```

"session_id",

"task_id",

"action_id",

"entry_type",

"timestamp",

"operator"

],

"validation_rules": {

"session_close_must_match_start": true,

"all_tool_calls_must_be_logged": true,

"memory_events_logged_when_occurring": true

}

}

Operational Discipline (Before Session 13 Begins)

Checkpoint	Verification Method	Owner
Ledger writable	Attempt test write, verify fallback works	Ken / Custody Server
Tools registered	claude mcp list confirms dossier-custody connected	Ken
Action counter recovered	Confirm starting A-number matches previous session end	Custody Server
Corpus integrity	Spot-check hash of 3 random files	Ken
Schema validated	LedgerWriter rejects invalid entry_type	Custody Server

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CROSS-MODEL ASSURANCE OBSERVATIONS

The pilot employs multiple AI models (Claude, Mistral, Grok) at different points. This introduces a **cross-model verification opportunity**:

1. **Claude Code** performs analysis
2. **Mistral/Grok** review findings independently
3. Disagreements flagged for human operator resolution

Assessment: This multi-model approach reduces single-agent bias but does not solve the ledger gap problem. Cross-model review validates *findings*, not *governance*. The ledger must still be fixed independently.

CONCLUSION

The Dossier Space Pilot architecture demonstrates a sophisticated understanding of the defensibility problem in AI-assisted evidence work. The separation of analysis from logging is correct and necessary. The pilot's documented failures are not evidence of design flaws but of **implementation gaps** that can be remedied.

Final Recommendation: Proceed to Session 13 with corrected ledger schema and mandatory completeness checks. Do not proceed with the current gaps — they create insurmountable audit deficiencies if challenged professionally.

The architecture itself is worth publishing. The failures, honestly disclosed, strengthen rather than weaken the governance narrative. Transparency about imperfection is itself a defensive asset.

APPENDIX A: LEDGER ENTRY TEMPLATE (RECOMMENDED)

```
{
 "ledger_entry_id": "<UUID v4>",
 "session_id": "DS-YYYYMMDD-S##",
 "task_id": "DS-YYYYMMDD-S##-T##",
 "action_id": "DS-YYYYMMDD-S##-A##",
 "entry_type": "<one of required_entry_types>",
 "timestamp": "ISO8601 UTC",
 "operator": "Ken Tombs",
 "description": "Human-readable summary",
```

```
 "material_decision": true | false,
 "severity": "critical | major | minor | informational",
 "payload": {
  "tool_name": "<if applicable>",
  "parameters": { ... },
  "result_set_count": <if applicable>,
  "gap_id": "<if applicable>",
  "context_window_size": "<if memory_compression>"
 },
 "status": "active | superseded",
 "superseded_by": "<ledger_entry_id if corrected>",
 "reconciliation_notes": ""
}
```

APPENDIX B: FINDINGS INTEGRITY VERIFICATION

Despite ledger gaps, the following findings remain independently verifiable:

Finding	Session	ROMER Record	Deliverable File	Hash Available
F-001 to F-004	S09.5	✓ Complete	✓ F-NNN-short-title.md	✓
F-005 to F-016	S12	✓ Complete	✓ F-NNN-short-title.md	✓
Supporting assets	All	✓ Complete	118 artifacts in GIT repo	✓

Conclusion: Findings integrity is preserved. Ledger gaps do not invalidate substantive outputs.

End of Review